

Multi-View Recognition of Driver Head Movements: A Measurement-First Study with Person-Held-Out Evaluation

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Abstract—We study four-class recognition of driver head-movement direction (left/right/down/up) from three synchronized cabin cameras, and report the results of a systematic 100+-run campaign on 22 drivers. Our contributions are threefold. (1) *Method*: a parameter-efficient multi-view architecture that keeps a single pretrained video transformer frozen and learns only a small cross-view token-attention module (11M trainable parameters). Under person-held-out evaluation it improves macro-F₁ by +6.3 points over an equally pretrained single-view model ($p=1.9 \times 10^{-4}$, paired bootstrap), and under a nested, selection-free protocol its micro-accuracy advantage grows to +5.1 points ($p=8.6 \times 10^{-12}$), while halving the across-driver variance. (2) *Measurement*: we show that two common protocol defects—a video-level split that leaks driver identity, and per-batch averaging of epoch metrics—jointly fabricated a coherent but false architecture narrative on this task; several “validated” conclusions, including our own earlier ones, reversed under a sound protocol. (3) *Negative results*: an analytic head-pose feature stream carries genuine signal in isolation, yet no injection route (token-, query-, or logit-level) survived robustness checks, and five class-imbalance remedies failed to prevent rare-class collapse. We release the person-held-out protocol, statistical-testing tooling, and all per-run predictions to support measurement-first research on small-cohort driver monitoring.

Index Terms—Driver monitoring, head movement recognition, multi-view fusion, video transformers, evaluation protocols, negative results.

I. INTRODUCTION

DRIVER-monitoring systems infer attention from head and gaze behaviour. We consider a cabin setup with three synchronized cameras (front, left, right) and the task of classifying short clips into four head-movement directions. The setting is representative of practical driver-monitoring research: a small cohort (22 drivers), heavy class imbalance, and strong inter-driver appearance variation.

This paper is deliberately *measurement-first*. Our initial campaign, run under a conventional protocol (random video-level split, standard metric logging), produced a seemingly well-supported progression of architectural improvements. Auditing the protocol invalidated essentially all of it. We therefore report (i) the method that survives a person-held-out, statistically tested evaluation; (ii) a controlled account of how the

defective protocol manufactured false conclusions; and (iii) the negative results of our attempts to exploit head-pose priors and to fix rare-class collapse.

II. TASK AND DATA

A. Recording and annotation

Each of 22 drivers was recorded in four illumination conditions (day/night $\times$ high/low), yielding 88 sessions. Three synchronized cameras (front, left, right; 332×224 px) observe the cabin. Timeline annotations mark head-movement segments with one of eight direction labels, merged to four (left/right/down/up; vertical components take precedence). Three annotators labelled every session; frame-level inter-annotator agreement is 91.0% overall and 94% for the *right* class, so label noise does not explain the failure modes reported below. Segment-level class frequencies are imbalanced: down 43%, left 28%, right 21%, up 9% (pooled over the five held-out folds). SAM-3D body keypoints (70 points, 2-D and 3-D) are available for every view.

B. Evaluation protocol

Person-held-out 5-fold CV. Drivers—not videos—are the unit of splitting: the 22 drivers are partitioned into five folds; each fold’s sessions form the evaluation set of a model trained on the remaining drivers. Pooling the folds scores every driver exactly once with a model that never saw them ($n=3954$ segments). **Nested variant.** For selection-free reporting, three additional drivers are carved out of each training set as an inner validation set; checkpoints are selected on the inner set and reported on the held-out fold, so the reported data never influences model selection. **Metrics.** micro-accuracy (segment accuracy), macro-accuracy, and macro-F₁, accumulated over the full epoch (Sec. VIII). **Statistics.** All model comparisons are paired on identical segment sets (verified fold-by-fold): McNemar’s test with continuity correction and a paired bootstrap (1,500–2,000 resamples, 95% CIs). The pooled binomial 95% CI is ± 1.6 points; a single 8-video split, by contrast, carries ± 6.4 points and up to 15.6 points of across-fold spread, which is why no single-split number appears in this paper.

C. Implementation

All arms train 50 epochs without early stopping (Adam, lr 10^{-4} , cosine schedule, effective batch 16, bf16), 32 uniformly

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Manuscript draft generated from the 2026-07 experimental campaign; co-author list and funding to be completed.

[Figure: three views $\rightarrow$ shared frozen ViViT $\rightarrow$ 17 tokens/view + view embeddings $\rightarrow$ cross-view transformer $\rightarrow$ mean pool $\rightarrow$ head]

Fig. 1. MV-Min: a frozen shared ViViT encoder with a small trainable cross-view token-attention module.

TABLE I
PERSON-HELD-OUT 5-FOLD CV (POOLED $n=3954$; SEED 42). “SPREAD” IS THE ACROSS-FOLD STANDARD DEVIATION OF MICRO-ACCURACY.

Model	micro-accuracy	mac. acc	macro-F ₁	spread
Majority class	42.7	25.0	15.0	.156
3D-CNN, single view	47.0	31.1	26.8	.134
TS-CVA mid fusion (3D-CNN)	45.4	31.9	28.1	.090
ViViT, single view	52.5	35.5	29.8	.134
ViViT, 3-view late fusion	52.6	35.7	31.9	.090
MV-Min (ours)	51.5	38.0	36.5	.090

sampled frames at 224^2 for ViViT arms, on one H100 per run. The video backbone is ViViT-B/16x2 pretrained on Kinetics-400. Class-frequency weights are applied to the cross-entropy loss in all arms.

III. METHOD

The method that survives evaluation is deliberately minimal (Fig. 1; “MV-Min” below). A *single, frozen* pretrained ViViT encoder processes each view. Per view, the 3137 output tokens are compressed to 17: the CLS token plus one spatially averaged token per temporal tubelet. A learned additive view embedding tags each view’s tokens, the three token sets are concatenated, and a 2-layer, 8-head transformer whose attention spans all views performs cross-view fusion. Mean pooling and a linear head produce the logits. Only the fusion module and head train (11M parameters, 4% of one finetuned backbone; the backbone runs under `no_grad`).

Design choices that our ablations showed to be unnecessary—partial backbone unfreezing, keypoint-trajectory tokens, keypoint-query pooling, gated aggregation—are excluded from the method and revisited as ablations in Sec. V and negative results in Sec. VI.

IV. MAIN RESULTS

Table I and the paired tests in Table II support three statements. **(1) Pretraining dominates micro-accuracy.** Replacing the 3D-CNN with a pretrained ViViT adds +5.5 micro-accuracy points; no fusion mechanism adds more on top of it. **(2) Cross-view attention buys balance and robustness, not raw accuracy.** Against the equally pretrained single-view

TABLE II
PAIRED COMPARISONS ON IDENTICAL SEGMENTS ($n=3954$). Δ IN POINTS WITH 95% BOOTSTRAP CI; McNEMAR p FOR MICRO-ACCURACY.

Comparison	Δ micro-accuracy	Δ macro-F ₁	p (McNemar)
MV full vs. 3D-CNN single	+5.1 [+3.6, +6.6]	+9.2	1.9×10^{-11}
MV full vs. ViViT single	−0.4 [−1.7, +0.9]	+6.3 [+4.8, +7.7]	0.55
MV full vs. ViViT late	−0.4 [−1.8, +0.9]	+4.1 [+2.4, +5.8]	0.55

TABLE III
COMPONENT ATTRIBUTION UNDER PERSON-HELD-OUT CV (PAIRED VS. THE ROW ABOVE IT OR AS INDICATED).

Arm	micro-accuracy	macro-F ₁	paired effect
MV-Min (frozen, no kpt)	51.5	36.5	reference
+ partial unfreeze (4)	52.4	36.2	n.s. / n.s.
+ kpt-query pooling	52.1	36.0	+1.3* / n.s.
+ unfreeze depth 6	50.5	35.5	n.s.
+ mirror augmentation	53.0	35.3	n.s.

*McNemar $p=0.004$; not robust under the nested protocol.

and trivial late-fusion controls, our fusion is micro-accuracy-neutral but gains +6.3 and +4.1 macro-F₁ respectively (both significant) and cuts the across-driver spread from .134 to .090. On a task whose practical failures are rare-class collapse and driver variance, this is the relevant axis. **(3) A previously “winning” fusion reverses.** TS-CVA mid fusion, the best fusion under our earlier leaky protocol (48.9 vs. 37.2 single-view), scores *below* the single-view 3D-CNN under person hold-out (45.4 vs. 47.0).

A. Nested, selection-free evaluation

Selecting checkpoints on the same people one reports on inflates results (measured: +2.0 points for the 3D-CNN arm, +0.9 for the multi-view arm). Under the nested protocol of Sec. II-B every number drops, but the ordering sharpens in our favour: MV-Min is the best arm (50.6 micro-accuracy), and the paired advantage of cross-view attention over the single-view control *grows* to +5.1 points ($p=8.6 \times 10^{-12}$); macro-F₁ +2.9 ([+1.4, +4.4]). The stricter the protocol, the clearer the method’s benefit—the opposite of what an overfit result would show.

B. Seed variance

Replicating both headline arms with seeds 42/43/44: single-view ViViT 52.5/53.0/51.9 micro-accuracy; the full multi-view arm 52.1/52.6/51.9. Run-to-run variation (± 0.6) is well inside the protocol CI, and the macro-F₁ advantage of cross-view fusion persists in all three seeds (mean +5.0).

V. ATTRIBUTION ABLATION

Table III: no addition to the minimal form produces a robust paired improvement on both metrics. Notably, conclusions that appeared firmly established under the leaky protocol—“query pooling beats token injection by +3.5”, “unfreeze depth 6 is optimal”, “partial unfreezing gains +2.9”—all shrink to non-significance or reverse. This motivates the minimal form as the published method.

TABLE IV

PER-CLASS F_1 UNDER PERSON-HELD-OUT CV: EVERY VIDEO MODEL COLLAPSES ONTO THE FREQUENT CLASSES; ONLY THE POSE-ONLY PROBE PREDICTS ALL FOUR.

Model	left	right	down	up
MV-Min	.538	.173	.673	.078
+ pose tokens	.532	.179	.695	.015
Pose-only probe (33-D)	.327	.237	.409	.117

TABLE V

REMEDIES FOR CLASS COLLAPSE, ALL EVALUATED UNDER THE SOUND PROTOCOL. NONE SUCCEEDED.

Remedy	micro-accuracy	outcome
Class-weighted CE (default)	51.5	collapse persists
Focal loss	−5.4 vs. ref	harmful
Balanced softmax	26.4	catastrophic
Clip-mean subtraction	43.6	harmful
Pose-token fusion	53.3 [†]	up- F_1 drops
Probe-model logit fusion (50/50)	44.1	harmful

[†]not robust (Sec. VI).

VI. HEAD-POSE PRIORS: SIGNAL WITHOUT A ROBUST INJECTION ROUTE

Head movements should be readable from head pose, and our scenario provides SAM-3D keypoints. A diagnosis of why naive keypoint streams failed is instructive: the raw 70×3 tensor is 60% hand rows (moving $4 \times$ more than the head), in unnormalized coordinates that encode driver identity, compressed into one token competing with 51 video tokens. The head signal never reached the model.

We therefore computed analytic features: 8 head-related points, a per-frame head-local frame from the ear axis and nose direction, yaw/pitch/roll plus first differences and centroid velocity plus local-frame shape (33-D/frame), with per-segment mean-pose subtraction to remove resting posture (identity).

The signal is real. A 64-unit MLP on these features alone—no video—reaches 27.2 macro- F_1 on unseen drivers with *non-zero F_1 on every class*: right 0.24 and up 0.12, where every video model scores ≈ 0.03 and 0.00 (Table IV).

No injection route survives. Per-frame pose tokens initially gave the best micro-accuracy of the campaign (53.3, $+1.7$ over MV-Min, $p=8.9 \times 10^{-5}$), but three robustness checks demoted it: (i) seed sensitivity ($+1.7/+3.3/+0.4$; significant in 2 of 3 seeds); (ii) *reversal* under the nested protocol (-2.5 , $p=3.2 \times 10^{-7}$)—part of the gain came from checkpoint selection on the scored drivers; (iii) a representation control showed raw flattened keypoint tokens capture most of the effect ($+1.1$, $p=0.021$), leaving $+0.6$ (n.s.) for the analytic encoding. Logit-level fusion of the pose probe with the video model, pre-registered as a 50/50 average, degraded micro-accuracy by 9.1 points ($p=5.4 \times 10^{-30}$): the probe is 22 points weaker overall, and the equal-weight blend loses more on frequent classes than it recovers on rare ones. We report these as negative results; the open problem is an injection mechanism that preserves the probe’s rare-class competence.

VII. RARE-CLASS COLLAPSE AND FAILED REMEDIES

Two persistent failure modes define the real difficulty of this task. **Collapse**: *right* and *up* are essentially never predicted by any video model (Table IV), despite 94% annotator agreement on *right*. Six remedies failed (Table V). **Driver memorisation**: under person hold-out the best checkpoint lands at epoch 0–4 in every arm—models start memorising training drivers almost immediately, consistent with the large across-driver spread. Removing static appearance (clip-mean subtraction) removed useful signal as well. We consider identity-invariant representations the key open problem for small-cohort driver monitoring.

VIII. HOW A DEFECTIVE PROTOCOL FABRICATED A NARRATIVE

Our first 40+ runs used (a) a random *video-level* split—all seven validation drivers also appeared in training—and (b) torchmetrics objects evaluated per batch (size 2) and averaged, which at that batch size inflates macro- F_1 by up to 17 points (0.52 logged vs. 0.35 true) and was inconsistently applied across trainer families. Together these produced a coherent, mutually confirming, and false progression (54.0 $\rightarrow$ 56.9 $\rightarrow$ 58.3 “SOTA” with supporting ablations). Under the sound protocol: the “+3.5 query-pooling” effect vanished; “optimal unfreeze depth” became n.s.; the TS-CVA fusion champion fell below its own single-view baseline; and an early “architectures are indistinguishable” reading—drawn from the same leaky split—also proved wrong. Single-split evaluation on 8 videos carries a ± 6.4 -point CI and 15.6 points of fold spread; every effect we chased for four rounds was smaller than that noise. We include this account because both defects are common, silent, and jointly self-reinforcing; the protocol of Sec. II-B is released as the antidote.

IX. CONCLUSION

Under person-held-out, statistically tested evaluation of 20+ arms (100+ trained models), the defensible findings are: (1) Kinetics-pretrained video transformers dominate micro-accuracy on this task ($+5.5$ over a 3D-CNN); (2) a frozen-backbone cross-view token-attention module—4% of the trainable parameters of one finetuned backbone—adds class balance ($+6.3$ macro- F_1), across-driver robustness (spread halved), and, under the strictest selection-free protocol, $+5.1$ micro-accuracy ($p=8.6 \times 10^{-12}$) over its single-view control; (3) head-pose priors carry real, complementary signal but lack a robust injection route; (4) rare-class collapse and driver memorisation, not architecture, are the binding constraints—six remedies failed; (5) two common measurement defects sufficed to fabricate an entire false research narrative on a small cohort. We release the protocol, tests, and per-run predictions.

REPRODUCIBILITY

All splits are person-disjoint and seeded; every comparison is paired on identical segments with McNemar and bootstrap statistics; per-run predictions are archived. Code, job scripts (90 runs), and the audit trail accompany the paper.

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